

BACHELOR'S THESIS

Exploring Hyperbolic Embeddings of Finite Metric Spaces

Raphael Vogt

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Supervisor: Prof. Dr. Christoph Schnörr

Faculty of Mathematics and Computer Science
Ruprecht-Karls-Universität Heidelberg

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Abstract

Abstract der Thesis

Zusammenfassung

Hier die Zusammenfassung

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1 Introduction

1.1 Overview and Motivation

This section gives an overview about prerequisites from math and computer science (complex numbers, vector spaces, gradients, data, automatic differentiation, representation learning), followed by a motivation about embeddings, why they are useful in practice. WENIG FORMELN, FOKUS AUF MOTIVATION UND INTUITIVES VERSTÄNDNIS.

1.2 Organization

This section describes how the following thesis is organized and which problems are tackled in which chapter.

2 Background

2.1 Riemannian Geometry

In order to gain an understanding of the embedding space, the basics of Riemannian geometry are now explained.

2.1.1 Tangent Space

In the following a basic understanding of smooth manifolds is assumed. This section follows the introduction of tangent spaces $T_p M$, $p \in M$ in chapter 1 of [Car92]. At every point p of an arbitrary smooth manifold M , tangent vectors can be defined. For this, $\alpha : (-\epsilon, \epsilon) \rightarrow M$ denotes a smooth curve which holds $\alpha(0) = p \in M$, i.e. it evaluates to p at $t = 0$. Additionally $\mathcal{D}$ refers to the set of functions, which are differentiable at point p . A tangent vector w at p is the tangent vector to $\dot{\alpha}(0)$ defined as

$$\dot{\alpha}(0)f = \left. \frac{d(f \circ \alpha)}{dt} \right|_{t=0}, \quad f \in \mathcal{D}. \quad (2.1)$$

The set of all tangent vectors spans the tangent space $T_p M$ at point p . Geometrically this space can be interpreted as the linear approximation of M at p .

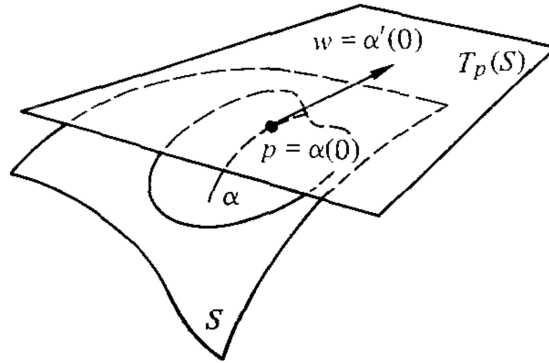


Figure 2.1: The tangent space of a surface S at point p [Car16]

2.1.2 Lengths and Metrics

While optimizing the embeddings, distances are computed in both the euclidean and the hyperbolic space. In the euclidean case, this is trivial using the euclidean metric, due to the flat space. In the hyperbolic case on the other hand, it's necessary to use some Riemannian metric. A Riemannian metric associates a smoothly varying symmetric and positive-definite bilinear form (inner product)

$$g(\cdot, \cdot) = \langle \cdot, \cdot \rangle_p : T_p M \times T_p M \rightarrow \mathbb{R} \quad (2.2)$$

to each point p on M [AMS09]. In practice, ds is also often used to denote the Riemannian metric. The Poincare's Disk model, which is used for the embedding space, can be considered a Riemannian manifold, which is a smooth manifold with an associated Riemannian metric [Car92]. Using the Riemannian metric of the manifold the length

$$L(\gamma) = \int_{t_0}^{t_1} \sqrt{g(\dot{\gamma}, \dot{\gamma})} dt \quad (2.3)$$

of a curve $\gamma : [t_0, t_1] \rightarrow M$ can be computed [AMS09]. The Riemannian distance applies this, to denote the distance between two points $x, y \in M$

$$\text{dist}(x, y) = \inf_{\Gamma} L(\gamma) \quad (2.4)$$

[AMS09] as the length of the shortest curve γ from the set Γ of all curves which pass x and y . Those length-minimizing shortest paths $\gamma(t)$ are called geodesics.

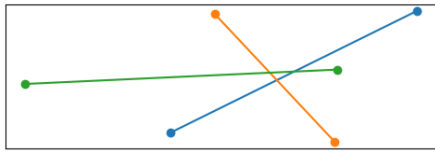


Figure 2.2: Geodesics in the euclidean plane

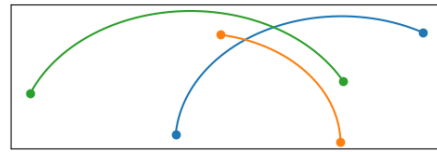


Figure 2.3: Geodesics in the hyperbolic Lobachevsky plane

2.1.3 The Exponential Map

During the optimization of the embeddings, data points in the embedding space are assigned to new points on the Riemannian manifold M . The direction and length of the

change depends on the Riemannian gradient $\text{grad } f$, which lies in the tangential space $T_p M$ of M at point p . The newly calculated value is assigned by the exponential map

$$\exp_p(v) := \gamma_{p,v}(1). \quad (2.5)$$

Geometrically speaking, it assigns a point $p \in M$ together with a tangent vector $v \in T_p M$ to a new point in M by evaluating the geodesic $\gamma(t)$ at $t = 1$ with initial velocity v at point p .

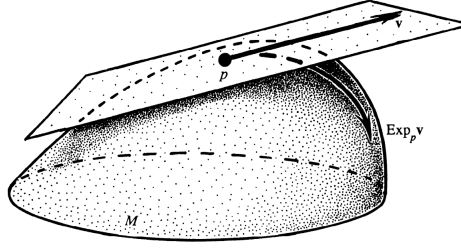


Figure 2.4: The exponential map at point p [Mor93]

2.2 Numerical Optimization on Manifolds

The embedding of the data follows an optimization approach. This means, instead of directly calculating good representations of the data in the embedding space, they are first randomly embedded and then continuously improved by minimizing a loss function $\mathcal{L}$. This chapter focuses on numerical optimization on manifolds in general, technical details will later be explained in chapter 3.2.

2.2.1 First Order Optimization

Before moving on to optimization on manifolds, consider the general optimization problem in the euclidean space

$$\inf_{x \in \mathbb{R}^n} f(x), \quad f \in \mathcal{F} \quad (2.6)$$

i.e. the objective function f from the set of functions $\mathcal{F}$ is getting minimized by finding optimal values of x . This can be achieved by moving x in the direction of the steepest descent $-\text{grad } f(x(t))$ until a critical point with $\text{grad } f(x(t)) = 0$ is reached. A numerical implementation is the gradient flow, which is the solution map $x_0 \rightarrow x(t)$ to the differential equation

$$\dot{x}(t) = -\text{grad } f(x(t)), \quad x(0) = x_0. \quad (2.7)$$

However, since this approach is impractical in most cases when using numerical methods [AMS09], the focus in the following is on algorithms

$$x_{t+1} = x_t + h_t \Delta x_t \quad (2.8)$$

which update x step wise in the search direction $\Delta x_t \in \mathbb{R}^n$ with step size $h_t \in \mathbb{R}$. In practice, this can be implemented algorithmically in the following way:

Algorithm 1 Basic optimization algorithm

Require: x_0

while stopping criterion not achieved **do**

1. Determine the search direction Δx_t

2. Find a suitable step size h_t

▷ Line search

3. Update $x_{t+1} = x_t + h_t \Delta x_t$

end while

More detailed information on the line search step can be found in chapter 2.2.3.

2.2.2 Riemannian Stochastic Gradient Descent

The following section explains the Riemannian stochastic gradient descent algorithm as described in [NK17] and [Bon13], which present an iterative algorithm for optimization on Riemannian manifolds. From now on $\theta_t \in M$ will be used to denote points on the manifold, η_t the learning rate at step t and $\mathcal{L}$ a loss function, which is the objective function that is getting minimized.

Algorithm 2 Riemannian stochastic gradient descent (RSGD)

Require: $\theta_0 \in M$, loss function $\mathcal{L}$

while stopping criterion not satisfied **do**

1. Compute $\text{grad } \mathcal{L}(\theta_t)$

2. Find a suitable step size η_t

▷ Line search

3. Update $\theta_{t+1} = \exp_{\theta_t}(-\eta_t \text{grad } \mathcal{L}(\theta_t))$

end while

In the algorithm, the Riemannian gradient of $\mathcal{L}$ is first computed. It is the tangent vector in $T_{\theta_t}M$ that describes the direction of the steepest ascent. It is then rescaled in

length with a suitable step size η_t and reversed, such that it points in the direction of the steepest descent. To obtain the new θ_{t+1} the exponential map is used. Information on how this procedure is used in practice for the Poincaré disk is discussed in chapter 3.

2.2.3 Line Search - Accelerating RSGD

For fast convergence in RSGD, it's necessary to choose a suitable learning rate η_t in each step t . To ensure this in every iteration, an adaptive learning rate is used during the optimization, which leads in practice to exponential convergence in the neighborhood of local minima [Sch23]. The simplest assumption how we can choose η_t is

$$\mathcal{L}(\exp_{\theta_t}(-\eta_t \text{grad } \mathcal{L}(\theta_t))) < \mathcal{L}(\theta_t) \quad (2.9)$$

such that the loss decreases in each iteration. However, this condition is not enough to obtain convergence [NW06]. Instead, a sufficient decrease condition is needed to guarantee convergence. A condition that ensures this is the Armijo condition

$$\mathcal{L}(\theta_t) - \mathcal{L}(\exp_{\theta_t}(-\eta_t \text{grad } \mathcal{L}(\theta_t))) \geq c \|\text{grad } \mathcal{L}(\theta_t)\|_g^2 \quad (2.10)$$

[Sch23] with $c > 0, \forall k \in \mathbb{N}$. Based on this condition, η is determined below in algorithm 3 using backtracking line search. This is an inexact line search algorithm that iteratively changes the step size until the Armijo condition is fulfilled. Inexact means that the optimal step size is not calculated, but an approximation that leads to a sufficient decrease, since calculating it exactly is too time-consuming in most cases.

Algorithm 3 Backtracking Line-Search

Require: $\theta_t \in M, \eta_t > 0, \tau, r \in (0, 1)$ (typical values : $\tau = \frac{1}{2}, r = 10^{-4}$)
while $\mathcal{L}(\theta_t) - \mathcal{L}(\exp_{\theta_t}(-\eta_t \text{grad } \mathcal{L}(\theta_t))) < r\eta_t \|\text{grad } \mathcal{L}(\theta_t)\|_g^2$ **do**
 Update step size $\eta_t = \tau\eta_t$
end while
return η_t

3 Poincaré Disk and Metric Data Embeddings

3.1 Hyperbolic Geometry

Hyperbolic geometry is a non-Euclidean geometry, which means that not all Euclidean axioms are fulfilled. To be precise, the fifth Euclidean axiom, which can be simpler expressed as the parallel postulate

”Given a line and a point not on it, there is exactly one line going through the given point that is parallel to the given line” [Can+97]

is not satisfied. Instead there can exist multiple lines through the given point which are parallel to the given line.

3.1.1 The Poincaré Disk

The model of hyperbolic geometry which will be used from now on is the Poincaré disk. It's the unit disk

$$\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\} \quad (3.1)$$

with the associated Riemannian metric

$$ds = \frac{2|dz|}{1 - |z|^2}. \quad (3.2)$$

This forms a model of hyperbolic space. The boundary of the disk, which is the sphere S^1 , is denoted as $\partial\mathbb{D}$. It represents points that are infinitely distant away from the center [Can+97].

Geometric properties

The Riemannian metric ensures that lengths grow exponentially with increasing distance from the center. It follows that geodesics are only straight lines if they are diameters, otherwise they are an arc of the circle that is orthogonal to $\partial\mathbb{D}$.

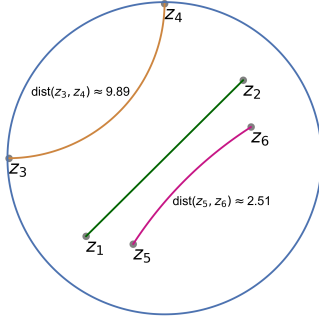


Figure 3.1: Geodesics in the Poincaré Disk

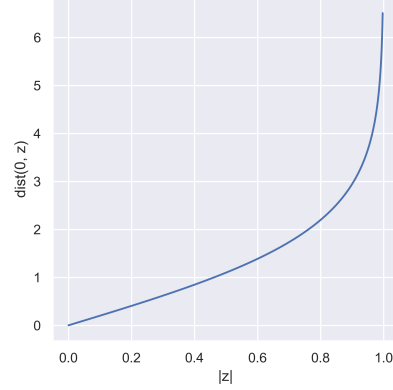


Figure 3.2: Growth of Riemannian distance in Poincaré Disk

3.1.2 Riemannian distances in the Poincaré Disk

The Riemannian distance between two arbitrary points $z_1, z_2 \in \mathbb{D}$ can be computed directly using the cross ratio. However this is hard in practice, because it requires to find the points where the geodesic between z_1 and z_2 intersects $\partial\mathbb{D}$. Instead, a method is now derived where z_1 and z_2 are distance preserving transformed inside the Poincaré Disk, such that z_1 gets mapped to the center of $\mathbb{D}$. Then the Riemannian distance between z_1 and z_2 simplifies to the distance from the center of $\mathbb{D}$ to z_2 which is easy to compute.

Möbius Mappings

A Möbius mapping is a Riemannian isometrie, which results in the fact, that Riemannian distances are preserved [Can+97]. In general, Möbius mappings can be represented as

$$f(z) = e^{i\theta} \frac{a + z}{1 + \bar{a}z}. \quad (3.3)$$

with $e^{i\theta} \in S^1$ and $a, z \in \mathbb{D}$.

Riemannian Distances in $\mathbb{D}$

To derive the distance formula, the Möbius mapping

$$f(z) = \frac{z - w}{1 - \overline{w}z} \quad (3.4)$$

is used. For two points $z, w \in \mathbb{D}$, it maps w to the center of $\mathbb{D}$ and thereby ensures that the geodesic connecting z and w becomes a diameter from the center in $\mathbb{D}$.

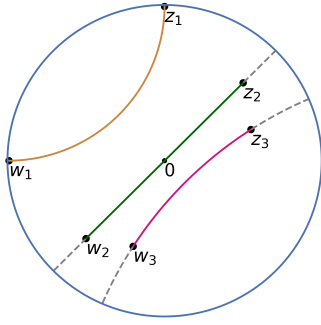


Figure 3.3: Geodesics in $\mathbb{D}$

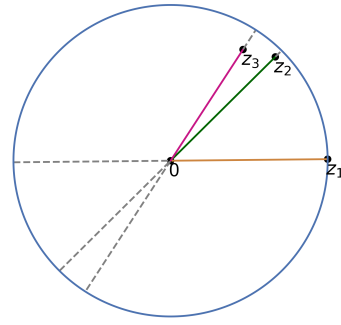


Figure 3.4: Geodesics after Möbius mapping

The length of the geodesic can now be calculated using the Riemannian metric ds . This leads to the Riemannian distance formula in $\mathbb{D}$

$$d(z_1, z_2) = d(f(z_1), f(z_2)) = d(0, f(z_2)) \quad (3.5)$$

$$= \int_0^{f(z_2)} \frac{2dt}{1-t^2} \quad (3.6)$$

$$= \log \frac{1+f(z_2)}{1-f(z_2)} \quad (3.7)$$

$$= \log \frac{|1 - z_1 \overline{z_2}| + |z_1 - z_2|}{|1 - z_1 \overline{z_2}| - |z_1 - z_2|} \quad (3.8)$$

between two arbitrary points $z_1, z_2 \in \mathbb{D}$ [And05].

Exponential Map on the Poincaré Disk

The exponential map on the Poincaré Disk for $z \in \mathbb{D}$ with the associated tangent vector $v \in \mathbb{C}$ is denoted as

$$\exp_z(v) = \frac{\lambda_z \left(\cosh(\lambda_z \|v\|) + \langle z, \frac{v}{\|v\|} \rangle \sinh(\lambda_z \|v\|) \right)}{1 + (\lambda_z - 1) \cosh(\lambda_z \|v\|) + \lambda_z \langle z, \frac{v}{\|v\|} \rangle \sinh(\lambda_z \|v\|)} z \quad (3.9)$$

$$+ \frac{\frac{1}{\|v\|} \sinh(\lambda_z \|v\|)}{1 + (\lambda_z - 1) \cosh(\lambda_z \|v\|) + \lambda_z \langle z, \frac{v}{\|v\|} \rangle \sinh(\lambda_z \|v\|)} v \quad (3.10)$$

[GBH18] with $\lambda_z := \frac{2}{1-\|z\|^2}$.

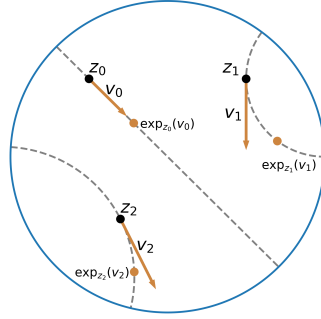


Figure 3.5: Exponential maps on the Poincaré Disk

3.2 Embedding Metric Data into the Poincaré Disk

3.2.1 Graph Data

The data that is embedded in the Poincaré disk has a graph structure. Formally a graph G is a tuple (V, E) consisting of a set of nodes (vertices) and a set of edges. This means that an edge $E \subseteq [V]^2$ represents the connection between two nodes in the graph. A path $P = (V, E)$ is a non-empty graph which shows the connection of two points in G . This leads to the distance $d_G(x, y)$ between two nodes $x, y \in G$, as the length of the shortest path connecting x and y [Die17].

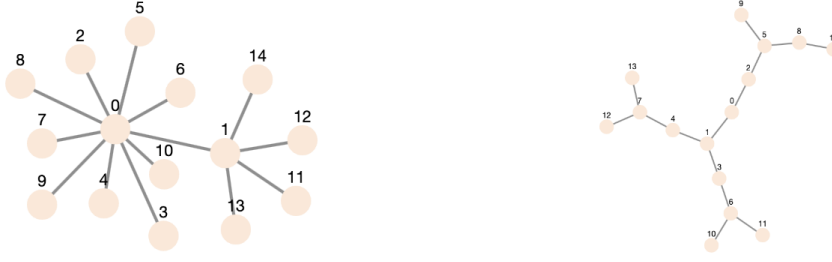


Figure 3.6: Two graphs with 15 nodes. The left has a high branching factor b and the right a low branching factor b

3.2.2 Loss Functions

Loss functions are used in the following as the objective function during the optimization process and to provide a way of measuring the quality of the embeddings. In the ongoing the distortion was chosen as the loss function. In general it consists of a combination of the pairwise distances in the input space and the embedding space. In the case of embedding graph data in the Poincaré Disk this results in the pairwise ratios

$$\rho(\theta_1, \theta_2) = \frac{d_{\mathbb{D}}(\theta_1, \theta_2)}{d_G(\theta_1, \theta_2)} \quad (3.11)$$

which uses the graph distance in the input space and the Riemannian distance in the embedding space [CL18]. There are many definitions of the distortion in the literature, one important which is later used in the experiments section is the average distortion

$$\Phi_{avg} = \frac{2}{n(n-1)} \sum_{\theta_1, \theta_2 \in \binom{\theta}{2}} \rho(\theta_1, \theta_2) \quad (3.12)$$

where $\binom{\theta}{2}$ denotes all pairs of data points [ABN11]. Further representations are provided, for example in [ABN11] (l_q -distortion) and [Cha+09] (ϵ -distortion).

Riemannian Gradient of Loss Functions

The Riemannian gradient of the loss function needs to be calculated in each optimization step. For that the Euclidean gradient of the loss

$$\nabla_E = \frac{\partial \mathcal{L}(\theta_1)}{\partial d(\theta_1, \theta_2)} \cdot \frac{d(\theta_1, \theta_2)}{\partial \theta_1} \quad (3.13)$$

is first calculated, which follows from the chain rule of calculus [NK17]. In practice, the gradient of $\mathcal{L}$ depends on the chosen representation of the distortion. The partial derivative of the Riemannian distance in the Poincaré disk results in

$$\frac{\partial d(\theta_1, \theta_2)}{\partial \theta_1} = \frac{4}{\beta \sqrt{\gamma^2 - 1}} \left(\frac{\|\theta_2\|^2 - 2\langle \theta_1, \theta_2 \rangle + 1}{\alpha^2} \theta_1 - \frac{\theta_2}{\alpha} \right) \quad (3.14)$$

with $\alpha = 1 - \|\theta_1\|^2$, $\beta = 1 - \|\theta_2\|^2$ and $\gamma = 1 + \frac{2}{\alpha\beta} \|\theta_1 - \theta_2\|^2$ [NK17]. Since the length of vectors in Euclidean space differs from those in the Poincaré disk, they have to be rescaled with the inverse of the Riemannian metric tensor

$$g_R = ds \cdot g_E = ds \cdot I_2, \quad g_R^{-1} = (ds \cdot I_2)^{-1} = I_2 \cdot ds^{-1} \quad (3.15)$$

of the Poincaré disk. This can be calculated trivially, because g_R results in a scalar matrix. This leads to the final representation

$$\text{grad } \mathcal{L}(\theta_t) = \frac{1 - |\theta_t|^2}{2} \nabla_E \quad (3.16)$$

of the Riemann gradient $\text{grad } \mathcal{L}(\theta)$ of the loss function.

3.2.3 Mini Batch RSGD

The RSGD algorithm presented in [algorithm 2](#) computes the gradient for a single data point, which results in a very noisy estimate of the gradient over the whole dataset. To get a better approximation of the gradient but still be tractable for large datasets, mini batch RSGD is used. In mini batch RSGD, the data is first split into random mini batches and then the gradient gets computed over the mini batch instead of only one data point [BB24]. In the algorithm θ_n represents the n -th embedding point and N the total number of data points.

Algorithm 4 Mini batch RSGD

Require: $\theta_{1:N} \in M$, loss function $\mathcal{L}$, batch size b

```

 $n \leftarrow 1$ 
while stopping criterion not satisfied do
  1. Sample mini batch  $B \leftarrow \theta_{n:n+b}$ 
  2. Compute  $\text{grad } \mathcal{L}(B)$ 
  3. Find a suitable step size  $\eta_t$  ▷ Line search
  4. Update  $\theta_{n:n+b} \leftarrow \exp_B(-\eta_t \text{grad } \mathcal{L}(B))$ 
  5.  $n \leftarrow n + b$ 
  if  $n > N$  then
    Shuffle data
     $n \leftarrow 1$ 
  end if
end while

```

Since mini batch RSGD introduces noise due to the repeated sampling of random mini batches, it is necessary to ensure that the learning rate decreases during the optimization, to obtain convergence [GBC16].

Optimal size of Mini Batches

4 Comparing Different Models of Hyperbolic Space - OPTIONAL

4.1 Existing Models of Hyperbolic Space

Poincaré half plane, Poincaré disk, Hyperboloid model, etc.

4.2 Evaluating the Models

5 Experiments and Discussion

5.1 Implementation Details

Implementation in torch or jax with focus on scalability and hardware acceleration

5.2 Evaluation on Toy and Real World Data

Results on toy data and real world data, correctness, distortion, runtime efficiency, memory efficiency

5.3 Conclusion

What went good, what wrong, what have we learned.

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